

Trustworthy LLM-Assisted Venture Screening: An Outcome-Free Measurement Audit

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Abstract

Early-stage investment screening requires experts to compare long, heterogeneous proposal documents containing narrative, tables, slides, technical claims, and financial assumptions. Large language models (LLMs) can reduce review effort, but a fluent report is not evidence that its scores are reliable. Conventional validation is also delayed: acquisition, failure, sustained revenue, and other venture outcomes may take years to observe and are strongly confounded by post-investment events.

We formulate LLM-assisted venture screening as a trustworthy-AI measurement problem. Rather than claiming to predict venture success, we audit whether LLM-generated scores behave as credible decision-support measurements before long-run outcomes are available. Our evidence-grounded pipeline extracts source-linked facts, constructs five investment-review dimensions, generates structured question-answer evidence, and maps the resulting report to an expert rubric. The audit evaluates construct alignment with experts, rank preservation, absolute-score bias, verdict calibration, claim grounding, and prompt-coupled response to controlled content perturbations.

On 12 expert-scored biotechnology, medical-technology, semiconductor, and related proposals, the system shows a fragile ordinal signal but fails several validity checks. Expert-rank alignment is moderate in this fitted sample (Spearman 0.589; pairwise concordance 0.759), but the rank correlation is only nominally significant and does not survive correction across the six reported rank tests. A direct raw-document baseline has lower Spearman (0.423) but slightly higher ICC (0.155 versus 0.119), and it compresses overall scores into only three distinct values. Scores are systematically shifted, dimension reliability is uneven, confidence is not validated, and QA-only scoring reverses expert ordering (Spearman -0.325). Rubric-targeted perturbations usually move scores in the expected direction, but those checks are prompt-coupled, nested within proposals, and do not repair weak expert agreement. **The evidence supports only audited reviewer triage, not autonomous investment judgment or venture-outcome prediction.**

CCS Concepts

• **Computing methodologies** → **Natural language processing**;
• **Information systems** → *Evaluation of retrieval results*; • **Applied computing** → *Economics*.

Keywords

trustworthy AI, large language models, venture screening, financial decision support, model validation, calibration, groundedness, multimodal documents

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1 Introduction

We study whether an LLM can support early-stage venture screening before investment outcomes are known, a setting where investors combine team, business, valuation, and post-investment considerations under uncertainty [13]. To make this question testable, we build an evidence-grounded pipeline that preserves proposal provenance, extracts atomic facts, organizes evidence into five investment dimensions, generates structured question-answer evidence, and produces rubric-aligned scores. We evaluate it on 12 expert-scored proposals. The central result is deliberately mixed: the system preserves some expert ordering, but absolute agreement is poor, scores are systematically shifted, and several dimensions fail to align with expert judgments. This paper therefore supports audited reviewer triage, not autonomous investment decisions or venture-success prediction.

The setting is important because investment committees compare proposals built on incomplete, mixed-format evidence. Empirical venture-capital research shows that deal selection spans sourcing, team assessment, business evaluation, valuation, contracting, and post-investment involvement, while ICAIF work frames innovation-finance support as a domain-specific classification and recommendation problem [13, 14]. A single submission may combine narrative claims, financial assumptions, technical diagrams, market tables, team biographies, intellectual-property statements, schedules, and regulatory plans; recent multimodal finance benchmarks show that such evidence requires reasoning across text, tables, figures, and domain knowledge [8]. LLMs are attractive for this workload, but financial benchmarks show that strong information extraction does not imply strong reasoning, forecasting, or decision performance [36].

Outcome-free auditing is not a substitute for eventual outcome validation; it is a proximal validity test. Acquisition, failure, sustained revenue, or a successful exit may occur years after screening and depend on follow-on capital, execution, market timing, regulation, macroeconomic shocks, and post-investment involvement [13]; VC-specific LLM benchmarks further emphasize sparse private signals, privacy constraints, and uncertain long-horizon labels [3]. Before those labels mature, a screening score should at least align with expert judgment, preserve relative priority, expose its evidence, calibrate its decision threshold, and respond to substantive content changes rather than presentation artifacts [12, 19]. Failure on these tests would make later predictive claims hard to trust; passing them does not prove venture success prediction, but it establishes whether the instrument is credible enough for constrained human review.

Recent ICAIF findings sharpen this audit design. FAITH shows that even when financial tables are supplied in context, LLMs can produce unsupported numerical answers, especially as reasoning moves from direct lookup to compositional table use [40]. The investment-bias study shows a different failure mode: models can preserve an initial investment preference even after receiving increasingly strong counter-evidence [20]. Saha et al. therefore argue that investment agents should be evaluated through intermediate behavior, robustness, and human collaboration rather than only final-task performance [32]. Following RAG/factuality work, we evaluate retrieval support and generated judgment separately, then compare the resulting measurements with expert scores and controlled content perturbations [10, 28].

The closest startup-evaluation system is DIALECTIC, which structures startup facts, question trees, pro/con reasoning, critique, and numeric investment scores [1]. Our novelty is not another multi-agent scoring architecture. We instead treat the produced score as the object of validation: does it reproduce expert ordering, use the numeric scale reliably, calibrate into screening verdicts, remain traceable to proposal evidence, and respond to controlled changes in substantive evidence? This measurement-validity framing turns startup evaluation from “can an LLM produce a plausible score?” into “what evidence would make that score trustworthy enough for human triage?”

Our central research question is:

Can an evidence-grounded LLM evaluation system serve as a credible outcome-free measurement instrument for venture proposal review?

This question decomposes into four empirical questions:

- (1) Does the system recover expert ordering and absolute scores?
- (2) Does calibration improve ranking and binary screening decisions?
- (3) Which investment dimensions remain unreliable?
- (4) Do scores move in the expected direction under rubric-targeted evidence additions and removals?

We make three contributions:

- (1) We implement a provenance-preserving venture-screening pipeline that separates source facts, dimension evidence, generated analysis, and rubric scores.
- (2) We operationalize outcome-free validation through expert agreement, pairwise ranking, absolute bias, post-hoc calibration, verdict behavior, claim grounding, and prompt-coupled content sensitivity.
- (3) We report a matched 12-proposal expert study showing a usable ordinal signal alongside scale bias, dimension-level unreliability, and failure modes that rule out autonomous deployment.

The paper’s claim is deliberately narrow: we evaluate whether the current system can assist reviewer triage under audit, not whether it can predict eventual venture success.

2 Related Work

2.1 Financial LLMs and Venture Screening

Financial LLM work has moved from general finance adaptation to task-specific evaluation. BloombergGPT, FinGPT, and PIXIU

show the value of financial pretraining, open data-centric adaptation, and instruction resources [35, 37, 38]; FinEval, CFBenchmark, and FinBen show that performance varies sharply across financial knowledge, reasoning, and decision-oriented tasks [22, 36, 39]. FinanceRAG and Multi-Reranker further show that retrieval, reranking, and context management materially affect grounded financial QA [11, 21]. These results motivate domain-specific evidence handling, but they do not test whether an investment-screening score behaves like a valid measurement of expert venture judgment.

Venture screening is a multi-construct expert task rather than generic text classification. Venture capitalists evaluate sourcing, team, business model, valuation, deal structure, and post-investment considerations, with deal selection central to value creation [13]. Recent LLM work targets adjacent objects: Automating Venture Capital and VCBench focus on founder or startup-success prediction under sparse private signals [3, 30], while DIALECTIC is closest architecturally because it structures startup facts, question trees, pro/con reasoning, critique, and numeric investment scores [1]. Our gap is not the absence of structured scoring; it is the absence of a matched measurement audit of whether full-proposal scores agree with experts, preserve rankings, calibrate verdicts, remain grounded, and respond to evidence changes before outcome labels mature.

2.2 Grounded and Trustworthy Financial Decision Support

Trustworthy financial AI requires calibration, robustness, traceability, uncertainty handling, and attention to false recommendations. ICAIF studies sharpen these requirements: investment-risk models can vary across country and gender attributes [2]; balanced arguments can reveal latent investment preferences that persist under counter-evidence [20]; and investment agents need intermediate-behavior and human-collaboration evaluation, not only final-task scores [32]. Broader human-AI decision studies likewise show that transparency, prompt design, trust, accountability, and task difficulty affect LLM-assisted choices [9, 23]. We therefore audit the behavior of the scoring instrument rather than the readability of its reports.

Grounded generation is equally central. RAGAS and ARES separate context quality from answer faithfulness [10, 31]; FActScore and SelfCheckGPT expose unsupported claims in long-form outputs [26, 28]; and FAITH shows that financial hallucination increases as table reasoning moves from lookup to compositional numerical inference [40]. Proposal review adds one complication: a defensible professional inference may be useful without appearing verbatim in the source. A finance-specific audit must therefore distinguish verifiable document claims from investment judgments.

2.3 Human-AI Agreement and Measurement Gap

Proposal review requires comparison with human reviewers rather than a fixed answer key alone. A systematic review warns that task formulation, metric choice, benchmark construction, and reporting can produce contradictory LLM-evaluation conclusions [19]. LLM-as-judge work supplies useful rubric and preference-evaluation

tools, but it also shows that the evaluator becomes part of the measurement apparatus: G-Eval uses criteria-guided evaluation [24], MT-Bench and Chatbot Arena expose preference biases [4, 41], and Prometheus-style models highlight the role of explicit rubrics [16, 17]. We therefore use human expert scores as the operational reference and report absolute agreement, rank ordering, calibration, and dimension reliability separately.

Taken together, prior work has advanced financial LLMs, VC-oriented prediction systems, structured startup scoring, grounded generation, and LLM-as-judge methods. The unfilled gap is narrower: no prior work, to our knowledge, audits a full-proposal LLM screening instrument against expert judgments while simultaneously testing rank preservation, scale bias, verdict calibration, evidence grounding, and pilot responsiveness before long-run venture outcomes are available. The following sections operationalize that gap as a measurement problem rather than as venture-success prediction.

3 Task Definition

Let D_i denote proposal i , represented as an ordered collection of pages or slides and their extracted evidence, and let H_i denote the corresponding expert assessment. The operational task is *screening*: produce an auditable assessment that helps reviewers prioritize proposals for further diligence. It is not a claim to estimate the probability of venture success.

Each proposal is evaluated along five ordinal dimensions. Every dimension score lies on the five-level expert rubric and is represented on $[0, 1]$ after normalization for analysis:

- **Team:** capability, governance, roles, and execution capacity.
- **Objectives:** clarity of goals, problem definition, milestones, and success indicators.
- **Strategy:** implementation path, market or deployment plan, and risk mitigation.
- **Advantages:** innovation, differentiation, intellectual property, and defensibility.
- **Feasibility:** resources, funding, schedule, technical practicality, and operational risk.

The expert rubric also contains an overall ranking score and a binary investment verdict. The system produces:

$$\hat{H}_i = \{\hat{h}_{i,\text{team}}, \hat{h}_{i,\text{objective}}, \hat{h}_{i,\text{strategy}}, \hat{h}_{i,\text{advantages}}, \hat{h}_{i,\text{feasibility}}, \hat{r}_i, \hat{c}_i, \hat{y}_i\},$$

where the first five outputs are normalized ordinal dimension scores, $\hat{r}_i \in [0, 1]$ is an overall screening score, $\hat{c}_i \in [0, 1]$ is the system’s confidence estimate, and $\hat{y}_i \in \{\text{GO}, \text{NO-GO}\}$ is a recommendation derived from calibrated score and confidence thresholds. Every generated factual claim is additionally associated with source evidence or marked as unsupported.

We adapt measurement-validity reasoning to this score-auditing setting: validity concerns whether evidence supports the intended interpretation and use of a score [27], and responsiveness to construct-relevant change is treated as a measurement property in applied measurement frameworks [29]. We separate four score-validity targets:

- (1) **Ordinal validity:** if experts prefer D_i to D_j , the system should usually produce $\hat{r}_i > \hat{r}_j$; we measure this through

Document upload → text/OCR extraction → atomic fact extraction
→ dimension construction → question generation → LLM answering
→ answer selection/post-processing → expert-style report → rubric scoring

Figure 1: Evidence-grounded proposal-screening pipeline. Each stage stores an auditable artifact rather than passing only free-form text.

- pairwise concordance and, for distance-sensitive monotonic association, Spearman correlation.
- (2) **Scale validity:** machine scores should not exhibit unexplained compression, offset, or threshold mismatch relative to the expert scale.
- (3) **Evidence validity:** source-checkable factual premises should be traceable to the submitted proposal. A related taxonomy requirement is that factual claims, professional inferences, and evaluator judgments should be distinguishable before groundedness is scored.
- (4) **Responsiveness validity:** scores should move in the expected direction under controlled, construct-relevant changes to the evidence.

The same targets apply at the dimension level by replacing $\hat{r}_i$ with $\hat{h}_{i,d}$ for each dimension d . Pipeline attribution is a separate architectural diagnostic rather than a score-validity target: it asks whether the evidence-grounded pipeline adds measurement value beyond raw-document scoring. Because long-run outcomes are unavailable, these targets provide an outcome-free audit of the instrument rather than external validation of venture success. The empirical section returns to the same vocabulary: RQ1 tests ordinal validity and the offset component of scale validity, RQ2 tests threshold calibration, RQ3 reports dimension reliability and score compression, the grounding pilot tests source-traceability after statement classification, and RQ4 reports a prompt-coupled pilot of responsiveness validity.

4 Evidence-Grounded Evaluation Pipeline

Figure 1 summarizes the implemented pipeline. Its objective is to preserve enough intermediate structure to locate failures in extraction, evidence organization, generation, or scoring.

This section describes the evaluated configuration, not independent proof that every stage is correct. The empirical audit in Sections 5–7 validates the end-to-end measurement behavior and selected failure modes; it includes a direct raw-document baseline but not stage-level ablations for extraction quality, question quality, or report generation. This distinction matters because later results show that sensible intermediate artifacts do not guarantee a valid final score.

4.1 Document Preparation

The system accepts PDF, DOCX, PPTX, TXT, and Markdown inputs. It extracts text while preserving page or slide boundaries, document identifiers, and source metadata. OCR is used for scanned or image-heavy pages; optional vision processing converts visible tables, charts, and diagrams into textual evidence. This reflects the

heterogeneous nature of financial documents [25], but the present evaluation does not isolate visual-reasoning accuracy. We therefore treat the evaluated representation as provenance-linked text.

4.2 Atomic Fact Extraction

The prepared corpus is chunked for LLM-based fact extraction. Each output record contains an atomic proposition, source location, evidence span, and tentative review dimension. The design attempts to keep facts separate from interpretations: “the proposal reports three patents” is a source claim, whereas “the intellectual-property position is defensible” is an evaluator judgment. This separation follows FActScore’s decomposition of long-form output into source-checkable atomic facts [28] and is essential for avoiding a category error in groundedness evaluation. It is also only a design constraint, not a solved problem: the pilot grounding audit later shows that an automated evaluator can still confuse generated judgments with source facts.

4.3 Dimension Construction

Extracted facts are assigned to team, objectives, strategy, advantages, and feasibility. For each dimension, the system constructs a provenance-linked evidence packet containing supporting facts, adverse facts, missing evidence, risks, and stated mitigations. Missing evidence is represented explicitly rather than silently filled by the model. These packets are the only proposal-specific context supplied to dimension-level question answering.

4.4 Question Generation and Answering

For each dimension, the system generates questions about rubric criteria, evidence sufficiency, risks, mitigations, and unresolved uncertainties. Answers use a structured schema that separates the direct response, factual claims, source pointers, confidence, caveats, and general domain knowledge. This separation follows RAGAS and ARES in treating context quality and answer faithfulness as distinct from response quality [10, 31]. General knowledge may inform interpretation, but it cannot be presented as evidence contained in the proposal.

4.5 Post-Processing and Report Generation

Candidate answers are checked for schema validity, dimensional relevance, evidence linkage, confidence, and internal consistency. Selected answers are compiled into an expert-style report with an executive summary, dimension-level findings, evidence gaps, scores, and a recommendation. The retained question-answer evidence lets reviewers inspect why a score was produced. This traceability is a core trustworthy-AI control: standardized finance benchmarks show that performance varies substantially across task types [36, 37], while finance-specific evaluation shows that plausible judgments can still contain systematic sensitivities [2].

4.6 Rubric Scoring

The final stage applies the same explicit rubric used for expert comparison. It produces five dimension scores, an overall screening score, a confidence value, and an initial recommendation. A post-hoc calibration layer then combines evidence-oriented QA scores with holistic investment-judgment scores using a blend weight λ ,

and maps continuous outputs to GO/NO-GO decisions. This blend is not a cosmetic implementation detail: Section 7 shows that QA-only scoring is negatively correlated with expert ranking in this sample, while investment-dominant weighting recovers positive ordinal alignment. The blend and verdict thresholds are therefore part of the measurement instrument being audited.

All proposals in the reported matched run use the same prompt templates, model identifier, decoding settings, and rubric. This fixes the evaluated configuration but does not make the pipeline strictly deterministic: the main dimension-analysis call uses nonzero temperature, and repeatability remains a deferred validation target.

4.7 Operational Footprint and Fallbacks

The completed 12-proposal OpenAI run used 1,868 logged LLM calls, or about 156 calls per proposal on average. The log records per-call latency (mean 8.74 seconds, median 7.46 seconds), but calls are partially parallelized, so summed call time is not wall-clock runtime. The audit artifacts do not record complete token usage or dollar cost for the full batch; the present paper therefore reports call volume rather than a reliable cost estimate. Any operational release should add per-stage token accounting and provider-price snapshots.

The workflow is not frictionless automation. Human intervention is required to supply proposal files, review failed or sparse OCR output, provide the expert score file, and inspect cases where provider calls fall back to local scoring or where grounding checks flag unsupported claims. The main fallback conditions are empty extraction, malformed JSON/schema outputs, provider failure, and grounding-category ambiguity. These fallbacks are part of the practical deployment boundary: the system can organize review evidence at scale, but it still requires analyst oversight when source extraction or evidence validation fails.

5 Evaluation Methodology

The evaluation follows the four validity targets from Section 3: ordinal validity through concordance and Spearman correlation, scale validity through ICC, score offset, threshold calibration, and compression, evidence validity through claim grounding, and responsiveness validity through controlled perturbation checks. The responsiveness experiment is prompt-coupled rather than independently generated, so it is treated as a pilot diagnostic. Uncompleted trustworthiness tests are identified separately.

5.1 Validation Targets

No single statistic establishes that an LLM score is trustworthy, so each target is evaluated with a small family of descriptive diagnostics. Human and AI scores are normalized to $[0, 1]$. Dimension scores are discretized to the nearest point on a five-bin grid $\{0, 0.25, 0.50, 0.75, 1.00\}$ for quadratic-weighted Cohen’s kappa, which gives partial credit for near misses and penalizes larger ordinal disagreements more heavily [5, 6]. Because kappa is sensitive to marginal bin occupancy, we report bootstrap 95% intervals only to show instability and imbalance, not as confirmatory tests against zero. We report:

- quadratic-weighted Cohen’s kappa for ordinal dimension agreement;

- intraclass correlation coefficient (ICC) and its 95% interval for absolute overall-score agreement, following standard reliability reporting guidance [18, 33];
- Spearman rank correlation for monotonic relationship [34];
- paired mean difference with a paired t -test and Wilcoxon signed-rank sensitivity check;
- pairwise rank concordance;
- verdict accuracy, precision, recall, F1, confusion counts, and exact binomial intervals where rates have small denominators.

Because Table 1 reports six Spearman correlations (five dimensions plus overall ranking), we treat them as a multiple-testing family. We report uncorrected values for transparency but interpret significance using Holm family-wise correction, which remains conservative while being less blunt than simple Bonferroni for correlated dimensions.

The five kappa estimates are not used as separate significance tests. Bootstrap intervals and bin counts are interpreted descriptively rather than as a family of accepted or rejected hypotheses.

Because scores are bounded in $[0, 1]$, paired t -tests are used as descriptive bias tests and checked against Wilcoxon signed-rank tests. The six paired mean-difference tests are treated as a second family under the same Holm correction. The overall score shift remains detectable (paired t unadjusted $p = 0.0015$, Holm-adjusted $p = 0.009$; Wilcoxon $p = 0.0024$), whereas the objective-dimension shift is directionally consistent but not confirmatory after family-wise correction (paired t unadjusted $p = 0.011$, Holm-adjusted $p = 0.057$; Wilcoxon $p = 0.027$).

5.2 Pairwise Rank Concordance

For each pair of proposals (D_i, D_j) where human rankings are not tied, we compare the human ordering with the AI ordering. Pairwise rank concordance is:

$$C = \frac{\text{\#concordant pairs}}{\text{\#comparable pairs}}.$$

This metric directly reflects screening: the system should preserve which proposal receives priority even if its numeric scale is shifted. It is related to Kendall’s pairwise formulation of rank correlation [15]. Human-tied pairs are excluded and AI ties on human-ordered pairs are counted separately.

5.3 Score and Verdict Calibration

For compactness, i indexes proposals and d indexes dimensions; q_{id} denotes the evidence-oriented QA score, v_{id} the holistic investment-judgment score, and $s_{id}(\lambda)$ their blend at weight λ . The proposal-level score and confidence are $\hat{r}_i$ and $\hat{c}_i$, with decision thresholds τ_r and τ_c . We audit the blend

$$s_{id}(\lambda) = \lambda v_{id} + (1 - \lambda) q_{id}, \quad \lambda \in [0, 1].$$

Using cached QA and investment scores, we sweep λ from 0 to 1 in increments of 0.05. This sweep diagnoses how score source changes rank agreement; it is not a held-out hyperparameter-selection procedure. The binary verdict is calibrated separately:

$$\hat{y}_i = \mathbb{K}[\hat{r}_i \geq \tau_r \vee \hat{c}_i \geq \tau_c],$$

The score and confidence thresholds are selected by a two-dimensional search over midpoints of the observed values. Because λ , τ_r , and τ_c are selected on the same 12 proposals used for reporting, calibration results are in-sample operating characteristics rather than generalization estimates.

Confidence estimation may use token probabilities, verbalized confidence, consistency, or hybrid signals, and calibration quality varies across tasks and elicitation methods [12]. Our reported confidence is therefore an empirical decision feature rather than a validated probability of correctness.

5.4 Claim Grounding

To evaluate whether generated reports are supported by the proposal text, we first classify report statements into four categories: verifiable facts, numerical inferences, professional inferences, and evaluator judgments. Only verifiable facts and numerical inferences enter the groundedness denominator. Professional inferences are instead treated as analyst judgments whose quality depends on the cited evidence and reasoning, while scores, confidence values, and GO/NO-GO recommendations are excluded because they are outputs of the evaluator rather than source-document claims.

For eligible factual or numerical claims, we assign one of three source-faithfulness labels:

- **supported**: explicitly stated in the proposal;
- **partially supported**: based on proposal evidence but incomplete, overgeneralized, approximate, or only partly justified;
- **unsupported**: absent from or contradicted by the proposal.

We report strict groundedness, half-credit lenient groundedness, and unsupported-claim rate. The strict/lenient distinction follows the observation that long-form generations mix directly supported facts, partially justified statements, defensible professional inferences, and unsupported claims [28]. Retrieval/context quality is separated from generation faithfulness following RAGAS and ARES [10, 31]. FAITH further shows why financial grounding must be conditioned on reasoning type: direct lookup and multi-step numerical inference have materially different hallucination profiles even when the relevant context is supplied [40]. Our audit generalizes the unit of analysis from masked numerical spans to factual claims in investment reports, while preserving a separate category for defensible professional inference.

$$G_{\text{strict}} = \frac{N_{\text{supported}}}{N_{\text{eligible}}},$$

$$G_{\text{lenient}} = \frac{N_{\text{supported}} + 0.5N_{\text{partial}}}{N_{\text{eligible}}},$$

$$U = \frac{N_{\text{unsupported}}}{N_{\text{eligible}}}.$$

5.5 Controlled Content Sensitivity

For strategy, objectives, and advantages, we construct proposal-specific paired PLUS and MINUS variants at the report/rubric input level. Each dimension contains eight fine-grained subrubrics. In the adaptive experiment reported here, a generation call at temperature 0 creates PLUS text covering all eight rubrics and identifies

exact proposal passages for MINUS replacement. The generator is shown the rubric anchors and asked to produce evidence that should raise the corresponding rubric scores; the MINUS generator is likewise asked to find evidence a scorer would use for high scores and replace it with absence statements. PLUS variants add concrete favorable evidence, such as a revenue model, regulatory pathway, intellectual-property status, benchmark result, market estimate, or buyer pathway. MINUS variants replace the strongest available evidence with explicit absence statements while preserving grammatical context.

Before rescore, each perturbation is assigned an expected direction and affected subrubric. A PLUS check is structurally testable only when its baseline score is below the maximum; a MINUS check is testable only when its baseline exceeds the minimum. Ceiling and floor cases are excluded rather than counted as failures. A remaining check passes when the targeted score moves in the expected direction. This design is related to ICAIF’s controlled evidence-conflict testing, which balances investment arguments and then increases counter-evidence to measure whether an LLM revises its prior view [20]. Our narrower test does not have the same independence property: because the generation prompt names the rubrics and desired scoring direction, the pass rate measures generation-scoring consistency under rubric-targeted edits, not independent evidentiary realism or freedom from confirmation, anchoring, or cross-rubric spillover.

The unit of intervention is nested: checks are grouped within proposals, dimensions, and subrubrics. We therefore report the directional pass proportion and use proposal-cluster bootstrap intervals as the uncertainty summary. We do not interpret the 417 checks as independent trials. We also report results by dimension and perturbation direction to expose heterogeneous behavior.

5.6 Deferred Trustworthiness Tests

Repeatability, rubric variants, style-only perturbations, cross-dimension halo tests, and cross-provider validation are part of the audit framework but not completed experiments. They correspond to the entries marked “Future” in Table 4 and remain necessary because LLM-as-judge work documents prompt, position, verbosity, and evaluator biases [4, 24, 41].

6 Experimental Setup

6.1 Dataset

The study contains 12 early-stage proposals from biotechnology, medical technology, healthcare AI, semiconductors, education technology, and related innovation domains. Inputs include PDFs and slide decks with Chinese and English text, scanned pages, tables, and technical figures. The mixed-language setting is relevant to Chinese financial benchmarks such as FinEval and CFBenchmark [22, 39]; however, the present sample is a venture-screening collection rather than a language benchmark.

The expert reference file contains 12 matched rows and no skipped rows. Each proposal has scores for team, objectives, strategy, advantages, feasibility, an overall 1–20 ranking, and a binary decision. The overall ranking is normalized to [0, 1] for comparison. The source proposals are confidential and should not be assumed releasable. The reproducibility artifact for this submission is therefore

limited to the schema, prompt templates, evaluation code, aggregate outputs, and anonymized metadata that do not expose proposal contents.

Statistical scope. All empirical results are pilot measurement-audit estimates over this 12-proposal matched sample. A Fisher-transformation power calculation [7] indicates that a two-sided correlation test at $n = 12$ needs approximately $|r| \approx 0.73$ for 80% power at $\alpha = 0.05$. The expert file is an operational reference rather than universal ground truth, calibration weights and verdict thresholds are selected on the same sample used for reporting, and the study has no held-out test set or second expert rater. We therefore use Holm correction, bootstrap intervals, bin counts, exact binomial intervals, and cluster bootstrap summaries to describe uncertainty, but we interpret the results as diagnostic evidence rather than deployable validation.

6.2 Evaluated Configuration

The reported run uses the nine-stage pipeline with OpenAI gpt-4o-mini and the investment-judgment weight $\lambda = 0.90$. Adaptive perturbation generation and subrubric rescore use temperature 0; the main dimension-analysis call uses temperature 0.25. The implementation can call other providers, but the matched results below should not be interpreted as cross-model evidence. The current experiment record includes the model family, provider, prompts, rubric text, run timestamp, aggregate evaluation outputs, and calibration parameters. It does not yet satisfy a full reproducibility checklist because prompt hashes, complete token accounting, retry counts, and fallback counts are not reported in the paper.

6.3 Compared Scoring Configurations

The present results contain three comparison families rather than a broad ablation suite:

- **Score-source sweep:** $\lambda = 0$ corresponds to QA-only scoring, $\lambda = 1$ to holistic investment-judgment scoring, and intermediate values blend the two cached scores. The reported $\lambda = 0.90$ is the operating point used for the main table.
- **Verdict-rule comparison:** the previous rule uses score ≥ 0.65 AND confidence ≥ 0.65 , whereas the recalibrated rule uses score ≥ 0.725 OR confidence ≥ 0.811 .
- **Direct raw-document baseline:** the same model scores each prepared `full_text.txt` document directly, without extracted fact packets, QA artifacts, or generated reports. The baseline uses temperature 0 and a 60k-character cap; all 12 proposal texts fit without truncation.

Removal of fact extraction, report-only scoring, and cross-provider comparison remain required follow-up experiments. FinanceRAG’s component ablations illustrate why these comparisons are necessary for attributing gains to pipeline stages [21]. The raw-document baseline is therefore a first attribution check, not a complete component ablation.

7 Empirical Measurement Audit

Table 1 reports human–AI alignment for all 12 matched proposals. We organize the results around the validity targets from Section 3: ordinal validity asks whether rankings are preserved, scale validity

Table 1: Human-AI agreement on 12 expert-scored proposals. Scores are normalized to 0–1; dimension agreement uses quadratic-weighted kappa with bootstrap 95% intervals, while overall ranking reports ICC and Spearman.

Dimension	<i>N</i>	Agreement	Spearman	Means H/AI
Team	12	0.324 [0.000, 0.550]	0.398	.729/.686
Objective	12	-0.138 [-0.333, 0.000]	-0.275	.833/.655
Strategy	12	-0.125 [-0.400, 0.111]	0.324	.708/.652
Advantages	12	-0.080 [-0.364, 0.020]	0.061	.667/.720
Feasibility	12	0.250 [0.000, 0.727]	0.156	.667/.683
Overall ranking	12	0.119 [-0.003, 0.257]	0.589	.816/.680

Agreement is quadratic-weighted kappa for dimensions and ICC for overall ranking.

Overall Spearman has uncorrected $p = 0.0437$ and Holm-adjusted $p = 0.262$ across the six Spearman tests, so it is treated as exploratory. Sparse five-bin occupancy makes kappa noisy: after snapping to $\{0, .25, .50, .75, 1\}$, four of five AI dimensions place 10 or 11 proposals in the 0.75 bin.

asks whether scores and thresholds match the expert scale, and evidence validity asks whether claims and judgments are properly supported.

7.1 RQ1: Ordinal and Scale Validity

Ordinal validity is partially supported, but scale validity is not. Table 1 shows moderate rank alignment but poor absolute agreement. The system correctly orders 44 of 58 expert-ordered pairs, yet the overall Spearman test does not survive Holm correction across the six reported rank tests. Its ICC is only 0.119 (95% CI $[-0.003, 0.257]$), and AI scores are lower than expert scores by 0.136 on average (paired t unadjusted $p = 0.0015$, Holm-adjusted $p = 0.009$ across six mean-difference tests). The defensible interpretation is narrow: the scores may help shortlist proposals, but they cannot substitute for expert numeric ratings.

The direct raw-document baseline does not close the validity gap, and it complicates the architecture story. Compared with one-shot raw-document scoring, the evidence-grounded pipeline has higher Spearman rank correlation (0.589 versus 0.423), but a paired bootstrap over proposals gives a wide interval for the difference ($\Delta = 0.167$, 95% CI $[-0.591, 0.711]$). The raw baseline has slightly higher ICC (0.155 versus 0.119), so the pipeline does not improve absolute agreement. Its more revealing failure is score compression: it produces only three distinct overall scores, differentiates only 17 of 58 expert-ordered pairs, and leaves 41 pairs tied. With the same OR threshold search, the raw baseline reaches 0.750 tuned verdict accuracy by predicting one GO proposal (TP=1, FN=3; exact 95% CI for accuracy $[0.428, 0.945]$). Structured evidence may improve differentiation, but it does not establish better absolute calibration or prove that each pipeline stage is necessary.

7.2 RQ2: Calibration and Decision Mapping

Calibration exposes a second scale-validity problem: pipeline outputs are not interchangeable. With $\lambda = 0$, the QA-only score is negatively correlated with expert rankings (Spearman -0.325 ; pairwise concordance 0.379). Increasing the weight on holistic investment judgment improves ordering. The key finding is that the

transformation from evidence-oriented QA scores to investment rankings is itself a model component that can fail.

The two-dimensional threshold grid shows how easily the apparent decision result changes. Predicting no proposals as GO yields 8/12 accuracy and zero recall; a strict one-GO threshold reaches 9/12 accuracy and 1/4 recall; the reported OR rule advances two proposals with 10/12 accuracy (exact 95% CI $[0.516, 0.979]$), 2/2 precision (CI $[0.158, 1.000]$), and 2/4 recall (CI $[0.068, 0.932]$). The previous AND rule advances nine proposals, recovers all four human GO cases, and creates five false positives (7/12 accuracy, 4/9 precision, 4/4 recall); the best grid point reaches 11/12 accuracy by advancing three proposals. We report the OR rule rather than the best grid point because the latter mainly illustrates threshold overfitting risk.

Confidence is also not yet a validated decision signal. It ranges from 0.633 to 0.818 and is negatively associated with absolute overall-score error, but the association is not statistically reliable (Spearman $r_s = -0.413$, $p = 0.183$). It is essentially unrelated to whether the binary verdict matches the expert verdict (Spearman $r_s = -0.048$, $p = 0.882$). Thus, the confidence term in the OR rule should be treated as an empirical feature, not as a calibrated probability of correctness.

7.3 RQ3: Dimension-Level Reliability

Dimension-level validity is heterogeneous, but low kappa is partly a scale-compression symptom. Team has the strongest quadratic-weighted kappa (0.324), followed by feasibility (0.250). Objective, strategy, and advantages have negative kappa. Objective is the clearest dimension-level scale-validity concern: experts assign a mean of 0.833, whereas the AI mean is 0.655, with an unadjusted paired t -test $p = 0.011$ and Wilcoxon signed-rank $p = 0.027$; after Holm correction across the six mean-difference tests, the paired t result is suggestive rather than confirmatory ($p = 0.057$). Advantages fails differently: the AI mean is slightly higher than the expert mean, but Spearman correlation is only 0.061. The model is therefore neither uniformly conservative nor uniformly aligned; its error mechanism depends on the construct being scored.

The kappa intervals and bin counts are as important as the point estimates. AI dimension scores concentrate heavily around the 0.75 bin while human scores occupy more of the upper and middle bins. Quadratic-weighted kappa is sensitive to this degenerate marginal distribution, so the negative values should not be interpreted only as substantive expert-model disagreement. They also diagnose that the AI rater is barely using the available ordinal scale. This compression is the range-restriction component of scale validity defined in Section 3, distinct from the offset and threshold-mismatch components tested in RQ1–RQ2. A screening instrument that assigns nearly all proposals the same dimension bin cannot support fine-grained triage, even if some individual judgments are plausible. Objective and advantages require direct expert review, and any user interface should expose dimension-specific reliability rather than implying that all sub-scores have equal validity.

Table 2: Corrected grounding audit after statement-type classification. Groundedness is scored only on verifiable facts and numerical inferences.

Metric	Value
Total classified report statements	188
Professional inferences excluded	114
Evaluator judgments excluded	41
Eligible factual/numerical claims	33
Supported claims	25
Partially supported claims	4
Unsupported claims	4
Strict groundedness	0.758 [0.590,0.872]
Half-credit lenient groundedness	0.818 [0.656,0.914]
Unsupported-claim rate	0.121 [0.048,0.273]
Proposals with eligible claims	11/12

7.4 Pilot Evidence-Validity Audit

Evidence validity is measurable but not yet strong enough for unattended use. Table 2 reports the corrected grounding audit. Across the 12 reports, the evaluator classified 188 substantive statements in the executive and expert-review sections. Most were professional inferences or evaluator judgments rather than direct source claims: 114 were professional inferences and 41 were evaluator judgments. The factual-grounding denominator therefore contains 33 verifiable facts and no numerical inferences. One proposal had no eligible factual or numerical claim in the evaluated sections after excluding scores, confidence values, verdicts, and professional judgments.

Of the 33 eligible claims, 25 are strictly supported, 4 are partially supported, and 4 are unsupported. The pooled strict groundedness is 0.758 (Wilson 95% CI [0.590, 0.872]), half-credit lenient groundedness is 0.818 (CI [0.656, 0.914]), and unsupported factual-claim rate is 0.121 (CI [0.048, 0.273]). A manual spot-check of all 33 eligible claims confirmed the need for taxonomy guardrails: score, confidence, verdict, and opinion-shaped sufficiency statements must be excluded before factual groundedness is computed. The audit therefore measures source traceability after statement classification; it does not independently validate taxonomy correctness. This corrected audit is informative, but it remains a pilot because the taxonomy and verification labels are model-assisted and not independently double-annotated.

7.5 RQ4: Prompt-Coupled Content Sensitivity

Table 3 reports directional checks for strategy, objectives, and advantages after excluding floor and ceiling cases. The scorer usually moves in the expected direction under rubric-targeted edits: 366 of 417 structurally testable checks pass (87.7%; proposal-cluster bootstrap 95% CI [83.3%, 91.7%]). Because the PLUS/MINUS generator was given the target rubrics and desired scoring direction, this is a prompt-coupled consistency result, not an independent test that realistic new evidence would cause the same movement. Objectives is most responsive (134/138; cluster CI [94.6%, 99.3%]), followed by strategy (137/151; cluster CI [87.0%, 94.6%]). Advantages is weaker (95/128; cluster CI [59.3%, 87.4%]), particularly when favorable evidence is added.

Table 3: Prompt-coupled content-sensitivity checks after excluding floor and ceiling cases. Counts are nested perturbations, not independent proposals.

Condition	Passed/Valid	Rate
Strategy PLUS	88/96	91.7%
Strategy MINUS	49/55	89.1%
Objectives PLUS	91/95	95.8%
Objectives MINUS	43/43	100.0%
Advantages PLUS	64/95	67.4%
Advantages MINUS	31/33	93.9%
Overall	366/417	87.7%

The perturbation results show that the rubric scorer reacts to edits engineered around its rubric, but they do not repair the weak expert agreement for objectives and advantages. Responsiveness and construct alignment are different properties: a score can move consistently under synthetic rubric-targeted changes while still using thresholds or interpretations that differ from experts.

7.6 Audit Coverage Summary

Table 4 summarizes the audit coverage after reporting the empirical results. This placement is intentional: the table is not a claim that the audit is complete, but a boundary around what the present paper establishes and what remains necessary before broader deployment.

8 Discussion

The results resist a simple “works/does not work” conclusion. The system contains a fragile ordinal signal and usually responds to rubric-targeted prompt-coupled edits. At the same time, its absolute scale is biased, several dimensions do not align with experts, and a naive component score reverses the ranking. This pattern gives empirical form to ICAIF’s broader warning that financial agents still lack standardized evaluation of intermediate behavior, robustness, and human collaboration [32]. Trustworthiness therefore comes from constraining the use case and exposing failure modes, not from treating the model as globally reliable.

The direct raw-document baseline complicates rather than confirms the architecture claim: one-shot scoring has weaker rank alignment and severe score compression, but its ICC is slightly higher than the pipeline’s. The honest interpretation is not that the pipeline dominates the baseline, but that structured evidence improves differentiation while absolute calibration remains unsolved. Likewise, the reversal from QA-only correlation of -0.325 to positive alignment under investment-dominant weighting shows that score construction and thresholding are part of the model, not book-keeping. Reviewers should inspect intermediate evidence and final scores separately because agreement at one layer does not imply agreement at another.

The narrow supported workflow is audited triage: use the model to organize heterogeneous evidence, flag missing information, and propose a preliminary ordering; retain expert authority over scores and decisions. Oversight should be dimension-specific, with stronger

Table 4: Outcome-free audit matrix for treating LLM proposal scores as measurement instruments.

Audit	Question	Failure mode	Primary metric	Status
Expert alignment	Do scores reproduce expert judgments?	The system measures a different construct.	Kappa, ICC, Spearman	Done
Rank preservation	Does the system preserve expert ordering?	Screening priorities are reversed.	Pairwise concordance	Done
Calibration	Are score and verdict mappings aligned?	Raw scores are compressed or threshold-misaligned.	Bias, accuracy, precision/recall	Done
Pipeline attribution	Does structured evidence improve over raw scoring?	Architecture adds complexity without measurement gain.	Direct baseline, ablations	Pilot
Content sensitivity	Do substantive additions/removals move the intended scores?	Scores ignore relevant evidence or only respond to rubric-targeted wording.	Prompt-coupled directional pass rate	Pilot
Groundedness	Are factual premises supported by proposal evidence?	Reports use hallucinated evidence.	Corrected groundedness, unsupported-claim rate	Pilot
Repeatability	Are identical runs stable?	Recommendations depend on sampling noise.	Score variance, verdict stability	Future
Style robustness	Does presentation alter scores without new evidence?	The system rewards polish rather than substance.	Score delta	Future
Halo/spillover	Are cross-dimension changes rubric-justified?	Salient traits contaminate unrelated scores.	Target/non-target movement	Future
Provider robustness	Do findings survive model changes?	Validity is provider-specific.	Cross-provider agreement	Future

warnings for objectives and advantages. Team scoring also creates fairness risk because founder names, biographies, affiliations, locations, nationalities, and gender-coded language may leak demographic or country signals into execution-capacity judgments; investment-risk work shows that LLM financial judgments can vary across country and gender attributes [2]. Any deployment should therefore mask or perturb demographic-adjacent cues and log when team judgments depend on affiliations rather than documented execution evidence.

Finally, proposal screening is a multimodal-document problem as well as a scoring problem. Source material spans pages, slides, OCR text, tables, and technical figures, and FinMR shows that expert-level financial multimodal reasoning still requires precise visual interpretation and domain knowledge [8]. Because we do not separately evaluate table understanding, OCR quality, or visual reasoning, the current results evaluate the end-to-end screening instrument rather than each modality. Confidential technology, financing, team, and intellectual-property material also make governance concrete: deployment must minimize retained text, restrict access to intermediate artifacts, log the prompt and model behind each recommendation, and support expert override.

9 Limitations

Beyond the statistical scope stated in Section 6, confidentiality limits release of source documents and makes independent reproduction harder. Mixed Chinese-English text, OCR, tables, and slides introduce extraction errors that are not separately quantified. The sensitivity experiment is explicitly prompt-coupled: the generation prompt names target rubrics and asks for changes expected

to move those rubric scores. PLUS passages may influence non-target rubrics, while MINUS replacements can leave semantically equivalent cues elsewhere in a report. The experiment therefore establishes generation–scoring consistency under rubric-targeted edits, not independent evidentiary realism or isolated causal effects for every subrubric.

The corrected grounding pilot separates source-checkable claims from professional judgments, but it is still not a final hallucination benchmark. Unlike FAITH’s answerable numerical spans, proposal assessments include professional inferences for which verbatim source matching is inappropriate [40]; the next grounding study should double-annotate the taxonomy and support labels with human reviewers. Repeatability, style-only robustness, halo/spillover, component ablations beyond the direct raw-document baseline, and cross-provider validation remain incomplete; these limitations correspond to the deferred audit categories in Table 4. A particularly important missing ablation is investment-judgment-only scoring over curated evidence (v_{id} alone) versus raw-document scoring: without it, we cannot isolate whether improvements come from evidence curation, investment-style prompting, or their interaction. The sensitivity experiment also does not test the balanced-conflict and escalating-counter-evidence conditions needed to detect confirmation bias [20]. LLM-based evaluation creates a second-order measurement problem: the judge can introduce its own prompt and preference biases [24, 41]. These limitations prevent claims of autonomous or broadly generalizable investment evaluation.

10 Conclusion

We presented an outcome-free measurement audit for LLM-assisted venture screening over heterogeneous proposal documents. The central finding is mixed: the system shows a fragile ordinal signal, prompt-coupled content sensitivity, and a corrected factual groundedness rate of 0.758 strict / 0.818 half-credit lenient over 33 eligible claims with wide intervals, but it does not reproduce the expert numeric scale, reliability varies sharply across dimensions, confidence is not yet validated, and QA-only scoring can reverse expert priorities. The direct raw-document baseline further clarifies the contribution: structured evidence improves score differentiation, but not absolute agreement.

The resulting deployment claim is intentionally narrow.

The system may support evidence organization, ranking, and analyst triage when calibration, provenance, and dimension-specific warnings are visible. The evidence does not support autonomous investment decisions or venture-outcome prediction. A larger held-out expert study, human-annotated claim-grounding evaluation, deeper component ablations, repeatability tests, and cross-provider analysis are required before operational deployment.

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